Omitted Context, Observable Mixtures, and Systematic Selection of False Complexity

Corrected theorem and streamlined proof

Revision of *Central theorem, end to end: from omitted context to systematic false-complex selection* (2 October 2026)

Abstract

We prove first, by an exact two-trial construction, that omitted context can make a homogeneous complex cognitive model strictly closer in joint Kullback–Leibler (KL) divergence to the observed data than a nested homogeneous simple model, even though both models are false. We then give a short general sufficient condition: a nonzero population extra-parameter score at the simple-model KL projection. With fixed numbers of trials and finite design support, BIC, Bayes factors and exact trial-level Bayesian leave-one-out (LOO) select the false complex model with probability tending to one. A separate local weighted-projection formula explains which design signatures generate the effect. The hierarchical extension is *conditional*: positive joint and conditional KL advantages must be established independently, and the regular local projection calculation requires interior, identifiable observable-law geometry. Zero-variance random-effects boundaries are not covered by that calculation. Relative to the initial draft, this version makes the closure formulation part of the definition of the risks and corrects score-drift, tangent-absorption, moment, parity and context-comparator overstatements; it also incorporates the fixes of an independent audit (29 October 2026).

1 Setting and the three distinct claims

Participants $i = 1, \dots, n$ are independent and identically distributed. A participant has a retained cell $W_i \in \mathcal{W}$ ($\mathcal{W}$ finite; $H(w) = \Pr(W = w) > 0$) and a fixed $T \geq 2$ binary-response vector $Y_i \in \{0,1\}^T$. The true conditional law $p(y \mid w)$ is positive in every cell. One motivating latent specification is

$$\Theta = a_0 + a_1 Z + \gamma C + U, \qquad \Pr(Y_t = 1 \mid \Theta = \theta, W = w, C) = \pi_t(\theta, w) = \operatorname{expit}\{\kappa_t(w) - \beta_t(w)\theta\}, \quad (1)$$

with conditionally independent trials given (Θ, W, C); C is omitted from the candidate fits. We write $\operatorname{expit}(x) = (1 + e^{-x})^{-1}$. The distribution of $\Theta \mid W = w$ is F_w, which can depend on U as well as C.

A candidate M is a set $\mathcal{Q}_M$ of conditional participant laws $q(\cdot \mid w)$, and $\mathcal{Q}_S \subseteq \mathcal{Q}_K$. The candidate families may be nonclosed; their closures are taken in the finite product of conditional probability simplices. Define

$$R_M = \min_{q \in \overline{\mathcal{Q}_M}} \mathrm{KL}_H(p \| q), \quad \mathrm{KL}_H(p \| q) = \sum_w H(w)\, \mathrm{KL}\{p(\cdot \mid w) \| q(\cdot \mid w)\}, \quad \Delta_J = R_S - R_K. \quad (2)$$

The minimum exists in the closure and is finite provided the family contains a positive law. As p has positive cells, every minimizing law $q_M^\star$ is positive. $R_M = 0$ *if and only if* $p \in \overline{\mathcal{Q}_M}$, not necessarily $p \in \mathcal{Q}_M$. Let $q_M^\star$ denote a unique optimal *law* when uniqueness is required; uniqueness of a parameterization is not assumed.

For the conditional target evaluated at the *joint* KL projections, let

$$C_M = \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{W, Y_{-t}}\, \mathrm{KL}\left\{p(Y_t \mid Y_{-t}, W) \,\|\, q_M^\star(Y_t \mid Y_{-t}, W)\right\}, \qquad \Delta_C = C_S - C_K. \quad (3)$$

This is exact *single-response* prediction: all other trials of the held-out participant remain observed. The joint projection is not refitted to minimize C_M. If the optimal law is not unique, C_M refers to a chosen optimal law; whenever C_M is used for LOO, uniqueness is assumed (Theorem 4.1).

Selection criteria. The data are n independent participant records $D = \{(W_i, Y_i)\}_{i=1}^n$; n counts participants, not trials. Each model is given by a measurable parameterization $\phi \mapsto q_M(\cdot \mid w; \phi) \in \mathcal{Q}_M$ with a fixed number k_M of free parameters and a fixed proper prior Π_M. With $\ell_{M,n}(\phi) = \sum_i \log q_M(Y_i \mid W_i; \phi)$, define

$$\mathrm{BIC}_M = -2 \sup_\phi \ell_{M,n}(\phi) + k_M \log n, \qquad m_M(D) = \int e^{\ell_{M,n}(\phi)}\, \Pi_M(d\phi), \qquad \mathrm{BF}_{KS} = \frac{m_K(D)}{m_S(D)},$$

$$\mathrm{ELPD}_M = \sum_{i=1}^n \sum_{t=1}^T \log \widehat{p}_M(y_{it} \mid D_{-it}), \qquad \mathrm{LOOIC}_M = -2\,\mathrm{ELPD}_M,$$

where D_{-it} deletes only the response y_{it} and $\widehat{p}_M$ is the exact posterior predictive (13). The supremum is used because maxima need not be attained. BIC, the Bayes factor and LOO *select* K when $\mathrm{BIC}_K < \mathrm{BIC}_S$, $\mathrm{BF}_{KS} > 1$ and $\mathrm{LOOIC}_K < \mathrm{LOOIC}_S$, respectively.

Three claims must be kept separate. (A) The exact witness and the homogeneous score-drift theorem are complete finite-support results. (B) The local weighted-projection formula is a regular, small-heterogeneity explanation, not a global equivalence. (C) Hierarchical selection requires distinct joint and conditional inequalities and, for the local formulas, additional geometric hypotheses.

2 Elementary facts about omission and product models

Lemma 2.1 (Mixture, conditional moments and covariance)**.** *Under* (1)*,*

$$p(y \mid w) = \int \prod_{t=1}^T \pi_t(\theta, w)^{y_t} \{1 - \pi_t(\theta, w)\}^{1-y_t}\, F_w(d\theta). \tag{4}$$

If conditional second moments exist and $m_C = \mathbb{E}(C \mid W)$*,* $m_U = \mathbb{E}(U \mid W)$*, then*

$$\mathbb{E}(\Theta \mid W) = a_0 + a_1 Z + \gamma m_C + m_U,$$
$$\mathrm{Var}(\Theta \mid W) = \gamma^2 \mathrm{Var}(C \mid W) + \mathrm{Var}(U \mid W) + 2\gamma \mathrm{Cov}(C, U \mid W).$$

If $\Theta \mid W = w$ *is nondegenerate and* $\pi_t(\cdot, w)$ *and* $\pi_u(\cdot, w)$ *are strictly monotone in the same direction,* $\mathrm{Cov}(Y_t, Y_u \mid w) > 0$*.*

Proof. Equation (4) is iterated expectation using conditional trial independence. The moment identities follow from linearity and the variance of $\gamma C + U$. For independent copies Θ, Θ' from F_w, conditional trial independence gives

$$\mathrm{Cov}(Y_t, Y_u \mid w) = \frac{1}{2}\mathbb{E}\big[\{\pi_t(\Theta) - \pi_t(\Theta')\}\{\pi_u(\Theta) - \pi_u(\Theta')\} \mid w\big].$$

The integrand is nonnegative and strictly positive with positive probability when both functions are strictly monotone in the same direction and F_w is nondegenerate. □

Write $p_t(\cdot \mid w)$ for the true trial marginal and $I_w = \mathrm{KL}\{p(\cdot \mid w) \| \prod_t p_t(\cdot \mid w)\}$ (multi-information).

Lemma 2.2 (Product reduction; the exact LOO identity)**.** *For every positive product law* $q(y \mid w) = \prod_t q_t(y_t \mid w)$*,*

$$\mathrm{KL}_H(p \| q) = \sum_w H(w) I_w + \sum_{w,t} H(w)\, \mathrm{KL}(p_t \| q_t), \tag{5}$$

$$\mathbb{E}_{W, Y_{-t}}\, \mathrm{KL}\{p(Y_t \mid Y_{-t}, W) \| q_t(Y_t \mid W)\} = \mathrm{I}(Y_t; Y_{-t} \mid W) + \sum_w H(w)\, \mathrm{KL}(p_t \| q_t). \tag{6}$$

Consequently, for two product-model families, $\Delta_C = \Delta_J / T$ *exactly. If the condition of Lemma 2.1 holds in a positive-probability cell, every product candidate satisfies* $R_M \geq \sum_w H(w) I_w > 0$*.*

Proof. Insert the product $\prod_t p_t(y_t \mid w)$ between p and q inside $\log(p/q)$ and average. For (6), insert p_t between $p(Y_t \mid Y_{-t}, W)$ and q_t. The dependence terms are independent of the candidate and cancel in $S - K$ differences. Positivity of covariance excludes a product truth, so the corresponding multi-information is positive. The inequalities extend to the closure because products form a closed subset of the finite simplex. □

Lemma 2.3 (What observing context buys)**.** *For any context-ignoring law $q(y \mid w)$,*

$$\mathbb{E}_{W,C}\,\mathrm{KL}\{p(Y \mid W, C) \| q(Y \mid W)\} = \mathrm{I}(Y; C \mid W) + \mathrm{KL}_H(p\|q). \tag{7}$$

Thus an exactly specified context-aware simple law has zero context-conditional *KL risk. A context-ignoring candidate has risk at least $\mathrm{I}(Y; C \mid W)$. Its optimal context-conditional risk is $R_K + \mathrm{I}(Y; C \mid W)$, which is positive if and only if $R_K > 0$ or the law of Y given (W, C) varies with C on a set of positive probability.*

Proof. Split $\log\{p(y \mid w, c)/q(y \mid w)\}$ into $\log\{p(y \mid w, c)/p(y \mid w)\} + \log\{p(y \mid w)/q(y \mid w)\}$ and average. □

Remark 2.4. Context omission need not create nonzero variability: dependence between C and U can cancel its contribution in (1). Conversely, heterogeneity in U alone can create the same mixture effect. Neither the variance identity nor the mixture identity, by itself, implies an advantage for K.

3 The exact constructive theorem (the shortest proof)

Theorem 3.1 (A false complex model wins consistently: exact two-trial witness)**.** *Take $T = 2$, $x_1 = 1$, $x_2 = 2$, $C \sim \mathrm{Bern}(1/2)$, $\Theta = C \log 3$, and conditionally independent responses with $\Pr(Y_t = 1 \mid \Theta) = \mathrm{expit}(-x_t\Theta)$. Compare homogeneous models*

$$S: \quad q_t = \mathrm{expit}(-x_t\theta), \qquad K: \quad q_t = \mathrm{expit}(\delta - x_t\theta),$$

with $\theta \in [0, 2]$ and $\delta = 4\{w(1/2; \rho) - 1/2\}$, $w(p; \rho) = \exp\{-(-\log p)^\rho\}$, $\rho \in [\rho_L, \rho_U]$, where $\rho_L = \log\{-\log(3/8)\}/\log\log 2 \simeq 0.0528$ and $\rho_U = \log\{-\log(5/8)\}/\log\log 2 \simeq 2.0600$ are the values giving $\delta = \mp 1/2$ (the project's minimal contract, also used in its pilot simulations); S is the restriction $\rho = 1$ ($\delta = 0$). Equip both models with proper priors having positive continuous density on their respective parameter boxes.

Then $0 < R_K < R_S$, the unique optimal laws exist, and $\Delta_C = \Delta_J/2 > 0$. As the number of participants increases with $T = 2$ fixed, BIC, the Bayes factor, and exact trial LOO all favor K with probability tending to one, although K is false. The extra parameter converges to $\rho_K^\star \neq 1$ under the usual likelihood or posterior law-consistency statement.

Proof. The true joint law, in the order $(00, 10, 01, 11)$, is exactly

$$(p_{00}, p_{10}, p_{01}, p_{11}) = \frac{1}{80}(37, 19, 13, 11). \tag{8}$$

Thus $p_1 = \Pr(Y_1 = 1) = 3/8$, $p_2 = \Pr(Y_2 = 1) = 3/10$, and

$$\mathrm{Cov}(Y_1, Y_2) = \frac{11}{80} - \frac{3}{8}\frac{3}{10} = \frac{1}{40} > 0.$$

Every product model is therefore false and has risk at least $I(Y_1; Y_2) > 0$ by Lemma 2.2. Model S cannot reproduce both trial marginals: matching them would demand

$$\mathrm{logit}(3/10) = 2\,\mathrm{logit}(3/8), \qquad \text{but} \qquad \log(3/7) \neq 2\log(3/5).$$

Model K *does* reproduce both marginals exactly with

$$\theta_K^\star = \log(7/5), \qquad \delta_K^\star = \log(21/25), \qquad \rho_K^\star \simeq 0.662697. \tag{9}$$

The corresponding ρ lies in (ρ_L, ρ_U) because $\delta_K^\star \in (-1/2, 1/2)$ and the Prelec map $\rho \mapsto \delta$ is strictly increasing. By (5),

$$R_K = I(Y_1; Y_2) > 0, \qquad R_S = I(Y_1; Y_2) + \min_\theta \sum_{t=1}^{2} \mathrm{KL}\{\mathrm{Bern}(p_t) \| \, \mathrm{Bern}(\mathrm{expit}(-x_t\theta))\} > R_K.$$

The strict minimum inequality follows by compactness and the impossibility of matching both marginals. Logit identifiability of the two distinct x_t gives a unique K law and unique δ and θ; strict convexity gives a unique simple projection. Lemma 2.2 yields $\Delta_C = \Delta_J/2$. The selection statement now follows from Theorem 4.1 below and prior support. Since the nested submodel has a larger risk, every optimal K law lies off $\delta = 0$, and the identified, continuous inverse from law to ρ gives parameter consistency. □

Independent numerical diagnostics	Value
Simple projection $\theta_S^\star$	0.4429045203
R_S; R_K (nats/participant)	0.0069680830; 0.0062695683
Δ_J; Δ_C	0.0006985147; 0.0003492573

Interpretive limitation. In this deliberately minimal design, the Prelec parameter is observationally a common offset, not separately identified as a unique psychological distortion mechanism. The theorem establishes false-*complexity* selection, not identification of probability weighting.

4 Why the selection criteria converge

Theorem 4.1 (Finite-cell selection transfer)**.** *Let $\mathcal{W}$ and the participant response space be finite, let $p(y \mid w) > 0$, and let each model contain a positive law. Suppose both model dimensions k_M are fixed and their proper priors put positive mass on every law-neighborhood of an optimal law. Nesting is not required. Then*

$$\frac{\mathrm{BIC}_K - \mathrm{BIC}_S}{n} \xrightarrow{p} -2\Delta_J, \tag{10}$$

$$\frac{1}{n} \log \mathrm{BF}_{KS} \xrightarrow{p} \Delta_J. \tag{11}$$

If, in addition, each closure has a unique optimal law $q_M^\star$, then exact single-response Bayesian LOO satisfies

$$\frac{\mathrm{LOOIC}_K - \mathrm{LOOIC}_S}{nT} \xrightarrow{p} -2\Delta_C. \tag{12}$$

Hence if $\Delta_J > 0$ and $\Delta_C > 0$, all three criteria select K with probability tending to one. If moreover $\mathcal{Q}_S \subseteq \mathcal{Q}_K$, the likelihood-ratio statistic for the nested restriction, $2(\sup \ell_{K,n} - \sup \ell_{S,n})$, diverges as $2n\Delta_J + o_p(n)$; any fixed critical-value test rejects with probability tending to one. This does not *assert a classical χ^2 null law under misspecification or irregularity.*

Proof. We outline the argument; Appendix A gives complete proofs (Lemmas A.1–A.4). Write $j = (w, y)$, $r_j = H(w)p(y \mid w) > 0$, and $\hat{r}_j$ for empirical participant-cell frequencies. Hoeffding's inequality and a union bound over the N cells give $\Pr(\max_j |\hat{r}_j - r_j| > u) \le 2Ne^{-2nu^2}$. Define $\ell(q) = \sum_j r_j \log q(y \mid w)$ and

$\hat{\ell}_n(q) = \sum_j \hat{r}_j \log q(y \mid w)$. Convergence of $\hat{\ell}_n$ to ℓ is uniform on laws whose cells are at least a fixed $a > 0$, and laws with a smaller cell are excluded from near-maximizers (Lemma A.1). It follows that

$$\sup_{q \in \mathcal{Q}_M} \hat{\ell}_n(q) \xrightarrow{p} \ell_M^\star = -R_M - \mathrm{Ent}_p(Y \mid W),$$

where $\mathrm{Ent}_p(Y \mid W)$ is the conditional entropy of Y given W. The BIC limit follows because $(k_K - k_S) \log n/n \to 0$. The evidence $m_M(D)$ is at most $\exp\{n \sup \hat{\ell}_n\}$. For every $\delta > 0$, a positive-prior-mass neighborhood of an optimal positive law has population log likelihood at least $\ell_M^\star - \delta$ and empirical log likelihood uniformly close to it; restricting the evidence integral to that neighborhood gives $n^{-1} \log m_M(D) \to_p \ell_M^\star$ (Lemma A.2). Subtraction yields the Bayes-factor limit.

For LOO, the exact predictive identity is

$$\widehat{p}_M(y_{it} \mid D_{-it}) = \frac{m_M(D)}{m_M(D_{-it})} = \int q_M(y_{it} \mid y_{i,-t}, W_i; \phi)\, \Pi_M(d\phi \mid D_{-it}). \tag{13}$$

The posterior is based on all full participant vectors except that Y_{it} is removed, and it uses the *marginal* law of the retained $Y_{i,-t}$. Because the optimal law is unique, there is a strict population likelihood gap outside each neighborhood of $q_M^\star$. A single empirical-cell frequency event controls *all* participant-deleted likelihoods: deleting any participant changes each empirical frequency by at most $1/(n-1)$, uniformly in i. The likelihood gap therefore gives exponentially vanishing posterior mass outside that neighborhood, uniformly over participant deletions. Restoring the other $T - 1$ responses of participant i reweights its participant-deleted posterior by the retained-response marginal $q_{M,-t}(y_{i,-t} \mid W_i; \phi) \le 1$. On a sufficiently small neighborhood of the positive optimal law, this reweighting has a uniform positive lower bound, so the conclusion is also uniform over single-response deletions (Lemma A.3).

Consequently, by continuity of conditional probabilities on the positive simplex,

$$\max_{i,t} |\widehat{p}_M(y_{it} \mid D_{-it}) - q_M^\star(y_{it} \mid y_{i,-t}, W_i)| \xrightarrow{p} 0.$$

The predicted probabilities are bounded away from zero with probability tending to one, so their logs converge uniformly. The participant-level law of large numbers applied to $T^{-1} \sum_t \log q_M^\star(Y_t \mid Y_{-t}, W)$ produces the LOO limit, because the true conditional entropies cancel in the $K - S$ difference (Lemma A.4). A union bound gives simultaneous selection. Under nesting, the likelihood-ratio assertion follows from the same likelihood limit. □

Remark 4.2. The proof works for hierarchical laws without treating a participant's trials as independent. It does not require a regular Laplace approximation to a marginal likelihood. The extra requirement for LOO is a *positive conditional gap*, not merely nesting or a positive joint gap.

5 General homogeneous theorem: the score-drift shortcut

A homogeneous model has $q_M(y \mid w; \phi) = \prod_t \mathrm{Bern}\{y_t; \mathrm{expit}(\eta_{Mt}(w; \phi))\}$. Suppose K embeds S at an extra parameter $\rho = \rho_0$ and $\phi_S^\star$ is an attained *interior* simple-model optimum. Let

$$e_t(w) = \left.\frac{\partial \eta_{Kt}(w; \phi_S^\star, \rho)}{\partial \rho}\right|_{\rho=\rho_0}, \quad \Lambda = \sum_{w,t} H(w) e_t(w) \{p_t(1 \mid w) - q_{St}^\star(1 \mid w)\}.$$

Theorem 5.1 (Strict false-complex advantage from extra-score drift)**.** *Assume the homogeneous setting above, with a differentiable feasible path in ρ through the simple optimum. If $\Lambda \neq 0$ and a small move in the direction*

of $\operatorname{sign}(\Lambda)$ *is permitted, then* $\Delta_J > 0$*. If in addition the truth is a mixture* (4) *with nondegenerate* F_w *and two same-direction strictly monotone trial kernels in some positive-mass cell, then*

$$0 < R_K < R_S, \qquad \Delta_C = \Delta_J/T > 0.$$

Under the support and unique-law conditions of Theorem 4.1, BIC, BF and exact trial LOO all select the false complex model with probability tending to one.

Proof. The population derivative of $\mathrm{KL}_H(p\|q_K(\phi_S^\star,\rho))$ at ρ_0 is $-\Lambda$. A feasible sufficiently small move in the improving direction decreases KL strictly below R_S, so $R_K < R_S$. Mixture-induced covariance gives $R_K > 0$ by Lemmas 2.1–2.2; the product reduction gives $\Delta_C = \Delta_J/T$. Apply Theorem 4.1. □

Remark 5.2 (Scope)**.** $\Lambda \neq 0$ is a *sufficient*, not necessary, condition for $\Delta_J > 0$. A zero derivative does not preclude improvement through second-order or nonlocal parameter changes. For example, on the design of Theorem 3.1 the family K' : $\operatorname{logit} q_t = -\rho^2 - x_t\theta$ ($\rho \in \mathbb{R}$) embeds S at $\rho = 0$ with $\partial_\rho \eta = 0$, so $\Lambda = 0$; yet K' contains every nonpositive offset, including $\delta_K^\star = \log(21/25)$, so $\Delta_J = 6.985 \times 10^{-4} > 0$. The same family illustrates the singular directions of Remark 6.3. Also, the displayed sum is a weighted score–residual inner product, not literally a covariance without centering or additional structure.

Proposition 5.3 (A useful global lower bound)**.** *Suppose the simple and complex logits are jointly affine in parameters* (ϕ, δ)*, the simple optimum is interior, and the nuisance information* $G_{SS} = J_S^\top \Omega^\star J_S$ *is positive definite, where* J_S *is the Jacobian of the simple logits with respect to* ϕ*, stacked over* (w,t)*, and* $\Omega^\star = \operatorname{diag}\{H(w)q_{St}^\star(1-q_{St}^\star)\}$*. Put* $e = \partial_\delta \eta_K$ *and let*

$$\Lambda_\delta = \sum_{w,t} H(w)e_t(w)\{p_t(1 \mid w) - q_{St}^\star(1 \mid w)\}$$

be the drift in the δ *coordinate (if* $\delta = \delta(\rho)$ *reparameterizes* ρ *near* ρ_0*, then* $\Lambda = \delta'(\rho_0)\Lambda_\delta$*). Let* $c = e - J_S G_{SS}^{-1} J_S^\top \Omega^\star e$ *and* $B = \frac{1}{4}\sum_{w,t} H(w)c_{wt}^2 > 0$*. If the affine path* $(\phi_S^\star - sG_{SS}^{-1}J_S^\top\Omega^\star e, \delta_0 + s)$ *remains feasible for every* s *between* 0 *and* Λ_δ/B *(automatic for a convex parameter domain once the endpoint is feasible), then*

$$\Delta_J \geq \frac{\Lambda_\delta^2}{2B}.$$

Proof. Along the path, the first KL derivative is $-\Lambda_\delta$ because the nuisance score vanishes at an interior optimum. The second derivative is $\sum_{w,t} H(w)\operatorname{expit}'(\eta_{wt}(s))c_{wt}^2 \leq B$. Taylor's inequality bounds the risk along the feasible segment above by $R_S - \Lambda_\delta s + Bs^2/2$; substitute $s = \Lambda_\delta/B$. □

6 A readable local mechanism: weighted residual alignment

Write the unobserved heterogeneity as $\Theta \mid w = \bar\theta(w) + \varepsilon V_w$, where $\mathbb{E}V_w = 0$, $\operatorname{Var}(V_w) = v(w)$, and $\sup_w \mathbb{E}|V_w|^3 < \infty$; all design quantities are fixed. Let $\pi_{t0} = \operatorname{expit}\{\kappa_t - \beta_t\bar\theta\}$, $\pi'_{t0} = \pi_{t0}(1-\pi_{t0})$ (the logistic derivative expressed through the probability) and $p_0(y \mid w) = \prod_t \operatorname{Bern}(y_t;\pi_{t0})$.

Lemma 6.1 (Attenuation and induced dependence)**.** *Uniformly on the finite* (w,y) *space,*

$$p_\varepsilon(y \mid w) = p_0(y \mid w)\{1 + \varepsilon^2(h_A + h_D) + O(\varepsilon^3)\}, \tag{14}$$

$$h_A = \sum_t a_t(w)(y_t - \pi_{t0}), \quad a_t(w) = \tfrac{1}{2}v(w)\beta_t(w)^2(1-2\pi_{t0}), \tag{15}$$

$$h_D = v(w)\sum_{t<u} \beta_t(w)\beta_u(w)(y_t - \pi_{t0})(y_u - \pi_{u0}). \tag{16}$$

Under p_0, h_A and h_D are centered and orthogonal. In particular,

$$\operatorname{logit}\Pr_{\varepsilon}(Y_t = 1 \mid w) = \operatorname{logit}\pi_{t0} + \varepsilon^2 a_t(w) + O(\varepsilon^3), \quad \operatorname{Cov}_{\varepsilon}(Y_t, Y_u \mid w) = \varepsilon^2 v \beta_t \beta_u \pi'_{t0}\pi'_{u0} + O(\varepsilon^3).$$

If V_w is symmetric with uniformly bounded fourth moment, the $O(\varepsilon^3)$ remainders above improve to $O(\varepsilon^4)$.

Proof. For $g_y(\theta) = \prod_t \operatorname{Bern}\{y_t; \operatorname{expit}(\kappa_t - \beta_t\theta)\}$, Taylor expansion and $\mathbb{E}V_w = 0$ give $\mathbb{E}g_y(\bar\theta + \varepsilon V_w) = g_y(\bar\theta) + \frac{1}{2}\varepsilon^2 v(w) g''_y(\bar\theta) + O(\varepsilon^3)$; bounded logistic derivatives and the moment assumption make the remainder uniform. Use $g''/g = \sum_t \ell''_t + (\sum_t \ell'_t)^2$, with $\ell'_t = -\beta_t(y_t - \pi_{t0})$ and $\ell''_t = -\beta_t^2\pi'_{t0}$. The binary identity $(y-\pi)^2 - \pi(1-\pi) = (1-2\pi)(y-\pi)$ gives (15)–(16). Centering and orthogonality follow from independence of the centered coordinates under p_0. Marginalization and the derivative of the logit give the displayed probability statements. Symmetry removes the cubic moment. □

The point is not that attenuation must lie outside S: it may be absorbed by a free intercept or another nuisance parameter. Only the *unabsorbed* component matters.

Theorem 6.2 (Homogeneous weighted-projection criterion)**.** *Suppose p_0 belongs to both homogeneous candidate families and their local logits have full-column-rank Jacobians J_S and $J_K = [J_S, e]$ at p_0. Assume the regular-chart hypotheses of Theorem 7.2 at p_0: embedded, locally isolated C^2 observable-law charts with p_0 interior and no other model-law branch approaching p_0 (automatically satisfied for full-rank affine-logit families at an interior parameter). Define*

$$\Omega = \operatorname{diag}\{H(w)\pi_{t0}(1-\pi_{t0})\}, \qquad M_M = I - J_M(J_M^\top \Omega J_M)^{-1} J_M^\top \Omega, \quad \tilde e = M_S e.$$

Then, with a given by (15),

$$R_M(\varepsilon) = \frac{\varepsilon^4}{2}\left\{\|M_M a\|_\Omega^2 + \|h_D\|_{p_0}^2\right\} + o(\varepsilon^4), \tag{17}$$

$$\Delta_J(\varepsilon) = \frac{\varepsilon^4}{2}\frac{\langle a, \tilde e\rangle_\Omega^2}{\langle \tilde e, \tilde e\rangle_\Omega} + o(\varepsilon^4), \tag{18}$$

where $\|h_D\|_{p_0}^2 = \sum_w H(w)v(w)^2 \sum_{t<u} \beta_t^2\beta_u^2\pi'_{t0}\pi'_{u0}$, and $\tilde e \neq 0$ because J_K has full column rank. If the charts are C^3 (for example, affine-logit families), both remainders are $O(\varepsilon^5)$. In particular, if $\langle a, \tilde e\rangle_\Omega \neq 0$, then $\Delta_J > 0$ for all sufficiently small nonzero ε and the local relative gain is determined by that partial alignment. For all homogeneous models, $\Delta_C = \Delta_J/T$ exactly.

Proof. Lemma 6.1 gives $p_\varepsilon = p_0\{1 + \varepsilon^2 h + O(\varepsilon^3)\}$ uniformly, with $h = h_A + h_D$ and $\mathbb{E}_0(h \mid W) = 0$. Theorem 7.2 below therefore applies with $k = 2$; its proof uses only finite-simplex compactness, the chart hypotheses and KL expansions, and does not depend on this section. It supplies the localization of the *global* projections at p_0, their uniqueness, and $R_M(\varepsilon) = \frac{1}{2}\varepsilon^4\|h - P_M h\|_{p_0}^2 + o(\varepsilon^4)$. It remains to compute the projections. A homogeneous model's scores are $\sum_t(\partial\eta_{Mt}/\partial\phi_j)(y_t - \pi_{t0})$, so $\mathcal{T}_M = \{\sum_t c_t(w)(y_t - \pi_{t0}) : c \in \operatorname{col} J_M\}$. Independence of the centered coordinates under p_0 gives $\langle\sum_t c_t(y_t - \pi_{t0}), \sum_t c'_t(y_t - \pi_{t0})\rangle_{p_0} = c^\top\Omega c'$, and h_D is orthogonal to every first-order function. Hence $P_M h = P_M h_A$ is the first-order function with coefficient vector $(I - M_M)a$, so $h - P_M h = \sum_t (M_M a)_t(y_t - \pi_{t0}) + h_D$ and $\|h - P_M h\|_{p_0}^2 = \|M_M a\|_\Omega^2 + \|h_D\|_{p_0}^2$; the formula for $\|h_D\|_{p_0}^2$ follows from independence. The weighted Frisch–Waugh–Lovell identity $\|M_S a\|_\Omega^2 - \|M_K a\|_\Omega^2 = \langle a, \tilde e\rangle_\Omega^2/\langle\tilde e, \tilde e\rangle_\Omega$ gives (18). For C^3 charts the minimizing chart coordinate is a C^2 function of the law (implicit-function theorem in the law variable, as in the proof of Theorem 7.2), so the fitted density ratio is $1 + \varepsilon^2 P_M h + O(\varepsilon^3)$ and both remainders are $O(\varepsilon^5)$. The final identity is Lemma 2.2. □

Remark 6.3 (Correct scope of absorption)**.** If $\tilde{e} = 0$, the additional parameter has no *first-order* observable direction at p_0 and formula (18) with division by $\langle \tilde{e}, \tilde{e} \rangle_\Omega$ is not applicable. Without further regularity, a locally redundant first derivative gives no reliable order-of-magnitude conclusion: a singular higher-order parameter direction can affect even the leading risk term. Under a regular, locally isolated equal-tangent-space setting, the regular projection theorem instead gives a vanishing leading-order gap. An *exact* zero gap for every positive truth requires that $\overline{\mathcal{Q}_S}$ and $\overline{\mathcal{Q}_K}$ contain the same positive laws, not merely equality of Jacobian spans at one point. Likewise $\langle a, \tilde{e} \rangle_\Omega = 0$ means only that the leading-order gain vanishes, not that $\Delta_J = 0$ for all ε.

7 Hierarchical models: a separate conditional extension

Hierarchical candidates are participant laws of the form $q_M(y \mid w; \phi) = \int \prod_t \mathrm{Bern}\{y_t; \pi_{Mt}(\vartheta, w; \phi)\}\, G_M(d\vartheta \mid w; \phi)$. Their trial predictions condition on the retained trials through the posterior distribution of the participant effect. The product reduction *does not apply*.

Theorem 7.1 (Hierarchical selection, conditional on two distinct gaps)**.** *Assume the finite-cell setting and prior support of Theorem 4.1, and unique optimal laws for exact trial LOO. For any nested hierarchical pair, if*

$$\boxed{R_S > R_K > 0 \quad \textit{and} \quad C_S > C_K,}$$

then BIC, the Bayes factor and exact single-trial LOO all select the false hierarchical complex model with probability tending to one. Neither the joint gap nor the conditional gap follows solely from omitting C. A nonzero feasible extra-score drift is sufficient for $R_S > R_K$, but the other two inequalities require separate proof.

Proof. Apply Theorem 4.1. The requirement $R_K > 0$ declares the selected model false. The drift implication is the same directional derivative argument as in Theorem 5.1, applied to the full participant likelihood rather than to product logits. □

Theorem 7.2 (Regular local hierarchical criterion, not a zero-variance theorem)**.** *Suppose $p_\varepsilon = p_0\{1 + \varepsilon^k h + o(\varepsilon^k)\}$ uniformly on finite cells, where p_0 is positive, $\mathbb{E}_0(h \mid W) = 0$, and $k \geq 1$. Assume both observable-law closures near p_0 are locally isolated, embedded C^2 manifolds with p_0 an* interior *point of each chart; their derivatives have full rank and tangent spaces $\mathcal{T}_S \subseteq \mathcal{T}_K$ in $L^2(Hp_0)$. Assume no other model-law branch approaches p_0 outside those local charts. Let P_M denote orthogonal projection, $r_M = h - P_M h$, $d = P_K h - P_S h$, and define the positive-semidefinite operator*

$$\mathcal{A} = \frac{1}{T} \sum_{t=1}^{T} \{I - \mathbb{E}_0(\cdot \mid Y_{-t}, W)\}.$$

Then unique optimal laws exist for all sufficiently small ε and

$$\begin{aligned} R_M(\varepsilon) &= \tfrac{1}{2}\varepsilon^{2k} \|r_M\|_{p_0}^2 + o(\varepsilon^{2k}), \\ \Delta_J(\varepsilon) &= \tfrac{1}{2}\varepsilon^{2k} \|d\|_{p_0}^2 + o(\varepsilon^{2k}), \\ \Delta_C(\varepsilon) &= \tfrac{1}{2}\varepsilon^{2k} \big\{ \langle d, \mathcal{A}d \rangle_{p_0} + 2\langle r_K, \mathcal{A}d \rangle_{p_0} \big\} + o(\varepsilon^{2k}). \end{aligned}$$

Therefore $d \neq 0$, $r_K \neq 0$, and $L := \langle d, \mathcal{A}d \rangle + 2\langle r_K, \mathcal{A}d \rangle > 0$ are sufficient for the three gaps of Theorem 7.1 for all small nonzero ε.

Proof. Finite-simplex compactness and the fact that $p_0 \in \mathcal{Q}_M$ force every global KL projection of p_ε to converge to p_0 in law space. Local isolation places it inside the specified chart. With full-rank derivative, the KL

Hessian at p_0 is positive definite in chart coordinates ψ, and by continuity $\psi \mapsto \mathrm{KL}_H(p\|q_M(\psi))$ is strictly convex on a fixed ball for every law p near p_0, so the minimizer in the chart is unique. Because p_ε need not be differentiable in ε, apply the implicit-function theorem with the *law* p as parameter: the stationarity equation $\nabla_\psi \mathrm{KL}_H(p\|q_M(\psi)) = -\sum_{w,y} H(w)p(y \mid w)\nabla_\psi \log q_M(y \mid w;\psi) = 0$ is linear in p and C^1 in ψ for a C^2 chart, so the minimizer $\psi_M(p)$ is C^1 in p near p_0. Since $p_\varepsilon - p_0 = O(\varepsilon^k)$, this and a second-order KL expansion give

$$q^\star_M/p_0 = 1 + \varepsilon^k P_M h + o(\varepsilon^k), \qquad \mathrm{KL}_H(p_\varepsilon\|q^\star_M) = \tfrac{1}{2}\varepsilon^{2k}\,\|h - P_M h\|^2 + o(\varepsilon^{2k}).$$

Nested orthogonal projections imply $r_S = r_K + d$ and $\langle r_K, d\rangle = 0$, yielding the joint gap. The KL chain rule for $Y_t \mid Y_{-t}, W$ subtracts the corresponding Y_{-t} marginal KL. Marginalizing a first-order density perturbation f replaces it with $\mathbb{E}_0(f \mid Y_{-t}, W)$, hence $C_M = \frac{1}{2}\varepsilon^{2k}\langle r_M, \mathcal{A} r_M\rangle + o(\varepsilon^{2k})$. Substitute $r_S = r_K + d$. Positive coefficients yield the stated strict inequalities. □

Remark 7.3 (Crucial boundary exclusion)**.** A Gaussian random-effect variance constrained to $\sigma^2 \geq 0$ is at a parameter-space boundary when $\sigma^2 = 0$. Ordinary interior-manifold hypotheses may fail there; tangent cones or direct one-sided arguments are needed. Theorem 7.2 does *not* prove the desired effect for a free-variance hierarchical model expanded around zero variance. Moreover L can be negative; $d \neq 0$ alone guarantees only the *joint* improvement.

Proposition 7.4 (Variance matching sharply limits the hierarchical gap)**.** *Assume* $\Theta \mid W = w = \bar\theta(w) + \varepsilon V_w$, $\mathbb{E}V_w = 0$*, and* $\mathrm{Var}(V_w) = v > 0$ *is constant in* w*. Suppose the hierarchical simple model can choose the correct means and a Gaussian shared effect with variance* $\varepsilon^2 v$*, and the trial product kernel is logistic with fixed finite design. If* $\sup_w \mathbb{E}|V_w|^3 < \infty$*, then* $R_S(\varepsilon) = O(\varepsilon^6)$*,* $0 \leq \Delta_J \leq R_S = O(\varepsilon^6)$ *and* $\Delta_C \leq C_S \leq R_S = O(\varepsilon^6)$*. If each* V_w *is symmetric and* $\sup_w \mathbb{E}|V_w|^4 < \infty$*, these bounds improve to* $O(\varepsilon^8)$*.*

Proof. Expand each positive participant-cell probability $\mathbb{E}g_y(\bar\theta + \varepsilon V_w)$ in ε. The Gaussian comparator matches the mean and variance terms through order ε^2, and bounded third absolute moments imply a uniform $O(\varepsilon^3)$ probability difference. Symmetry plus bounded fourth moments improve this difference to $O(\varepsilon^4)$. Since every baseline cell is strictly positive, KL between normalized nearby laws is quadratic in their probability difference, giving $O(\varepsilon^6)$ or $O(\varepsilon^8)$. The optimal simple risk cannot exceed the comparator risk, and nesting implies $0 \leq \Delta_J \leq R_S$. Finally, by the chain rule $C_M = T^{-1}\sum_t\{\mathrm{KL}_H(p\|q^\star_M) - \mathrm{KL}_H(p_{-t}\|q^\star_{M,-t})\} \leq R_M$, and $C_K \geq 0$, so $\Delta_C \leq C_S \leq R_S$. □

Proposition 7.5 (Corrected symmetry/parity obstruction)**.** *Let* $Z \in \{-1, 1\}$ *be balanced, with a common trial design and an otherwise* Z*-symmetric* interior positive-variance *baseline law. Suppose the true local perturbation is* λh *with* $h(Z, y) = Z s_v(y)$*, the simple tangent space* $\mathcal{T}_S$ *is invariant under the reflection* $Z \mapsto -Z$*, i.e. it is the direct sum of its even and odd parts (e.g., the scores for* μ_0, v *are even and that for* μ_1 *is odd), and* the additional complex-model score is Z-even at the baseline. *Under the regular-chart hypotheses of Theorem 7.2,* $P_K h = P_S h$ *and* $\Delta_J(\lambda) = o(\lambda^2)$*. If* $h = Z s_v$ *has a component orthogonal to the odd part of* $\mathcal{T}_S$*, then* $R_S(\lambda)$ *is positive to order* λ^2*.*

Proof. Balanced Z makes the even and odd subspaces orthogonal in $L^2(Hp_0)$. The perturbation is odd and the added score is even. Because $\mathcal{T}_S$ is reflection-invariant, residualizing the added even score against $\mathcal{T}_S$ leaves an even function, which has zero inner product with the odd perturbation. Thus adding it does not enlarge the projection of h, and the first-order-in-λ version of Theorem 7.2 yields the claims. □

Remark 7.6. "Adds one global parameter" is not a sufficient parity assumption. An extra global parameter entering as $v(Z) = v + \delta Z$ has an *odd* score and may fit context-dependent variance exactly. That counterexample is excluded by the explicit even-score hypothesis above.

Proposition 7.7 (Joint KL improvement need not imply conditional improvement)**.** *Even for nested, positive finite-dimensional participant-law families there can be $\Delta_J > 0 > \Delta_C$. For $T = 2$, in the cell order $(y_1y_2) = (00, 10, 01, 11)$ of Theorem 3.1, take*

$$p = (.27, .33, .28, .12), \quad s = (.11, .19, .12, .58), \quad k = (.68, .06, .05, .21),$$

and $\mathcal{Q}_S = \{s\}$, $\mathcal{Q}_K = \{s + \lambda(k - s) : 0 \le \lambda \le 1\}$. The unique joint projection onto $\mathcal{Q}_K$ has $\lambda^\star \simeq 0.281900$, and

$$\Delta_J \simeq 0.098320 > 0, \qquad \Delta_C \simeq -0.052440 < 0.$$

Thus neither nesting nor a joint likelihood advantage establishes an exact trial-LOO advantage for general dependent-response models.

Proof. All displayed cells are positive. The function $\lambda \mapsto \mathrm{KL}\{p\|s + \lambda(k - s)\}$ is strictly convex, with derivative $-\sum_j p_j(k_j - s_j)/(s_j + \lambda(k_j - s_j))$. This derivative has a unique zero in $(0, 1)$, giving the stated $\lambda^\star$. Compute $C(q) = \mathrm{KL}(p\|q) - \frac{1}{2}\{\mathrm{KL}(p_{-1}\|q_{-1}) + \mathrm{KL}(p_{-2}\|q_{-2})\}$ for $q = s$ and $q = s + \lambda^\star(k - s)$ to obtain the displayed gaps. The comparison is an abstract nested law-space example, *not* a constructed witness for a particular Gaussian cognitive random-effects pair. □

8 Context-aware comparison and theorem status

Corollary 8.1 (Measured context removes the correctly modeled component)**.** *Let C have finite support and be observed; suppose the simple context-aware model S_C contains the true law $p(Y \mid W, C)$, including any residual U distribution required for correctness. Compare it with a context-ignoring K evaluated on (W, C), using the same participant-level likelihood and fixed-dimensional proper priors. If $R_K + \mathrm{I}(Y; C \mid W) > 0$, then the* context-conditional *joint KL gap favoring S_C is strictly positive. BIC and BF consequently favor S_C with probability tending to one. If optimal laws are unique, exact single-trial LOO also favors S_C with probability tending to one.*

Proof. The context-aware truth has risk zero. Identity (7) makes the context-ignoring model's risk $R_K + \mathrm{I}(Y; C \mid W) > 0$. For any two positive joint laws, zero conditional KL for every $Y_t \mid Y_{-t}, W, C$ implies equality of all full conditionals and hence equality of joint laws (ratios of their masses are constant along edges of the binary hypercube). Therefore a positive joint KL risk implies a strictly positive average conditional KL risk when comparing the true law to the optimal incorrect law. Apply Theorem 4.1, which does not require nesting, on the enlarged finite design (W, C) with S_C in the role of the preferred model. □

Remark 8.2. The positive-gap assumption matters: if K also represents the truth on (W, C), an asymptotic n-order preference cannot be inferred from KL alone. Including C also does not make the simple model correct if the remaining U heterogeneity is not modeled adequately.

8.1 What is proved, and what is not

Regime	Mathematical status in this revision
Exact witness	Fully proved within stated parameter boxes and priors; nonzero joint and trial-conditional gaps calculated explicitly.
General homogeneous	Fully proved *conditional on the explicitly checkable score drift*, nondegenerate mixture response, and finite-cell selection assumptions.
Local homogeneous	Leading-order theorem under regular local charts and full-rank logit signatures; not a global necessary-and-sufficient condition.
Hierarchical selection	Complete *transfer theorem* when $R_S > R_K > 0$ and $C_S > C_K$ have separately been established. These signs are not proved for an arbitrary cognitive hierarchical pair.
Hierarchical perturbation	Valid under the interior isolated-manifold assumptions. It does not cover free random-effect variance converging to zero without additional one-sided geometry.
Hierarchical existence witness	*Open for the specific G1 pair.* No new empirical context/design construction verifying all hierarchical signs is claimed here.

8.2 Corrections to the 2 October draft

1. **Closure and attainment:** Original Lemma 2.1 already stated attainment on $\overline{\mathcal{Q}_M}$ and $R_M > 0 \iff p \notin \overline{\mathcal{Q}_M}$. This version makes the closure part of the definition of R_M and of the optimal laws used throughout.
2. **Marginal shape:** Original Corollary 1.5 does not imply that every mixture's marginals escape S; only their unabsorbed design component matters.
3. **Drift:** Original Theorem 2.4 was a one-way implication, but its first remark overstated it ("occurs exactly when"). That remark is dropped; $\Lambda = 0$ may coexist with improvement at higher order or elsewhere (Remark 5.2).
4. **Gap bound:** Original Proposition 2.5 assumed an interior projection and a feasible step. This version also states positive-definite nuisance information, a positive curvature bound and feasibility of the whole path segment.
5. **Regular local geometry:** Original Theorem 2.6 becomes an explicitly interior conditional theorem. No zero-variance Gaussian boundary is silently treated as regular.
6. **Tangent absorption:** Original Corollary 2.7 and Proposition 4.2 distinguish zero leading-order improvement from exact equality of candidate law spaces.
7. **Higher moments:** The third-moment condition of original Theorem 1.4, and for the improved rate symmetry and fourth moments, are now stated in the proposition itself rather than inherited.
8. **Parity:** Original Proposition 4.4 now requires the extra complex-model score to be Z-even.
9. **Trial LOO:** The separate conditional-gap condition is retained and accompanied by an explicit nested-law sign reversal, not merely a reference to a separate file.
10. **Context comparator:** The context-aware result now states correct specification of the residual law, positive context-conditional gap and the required unique-law condition for LOO.
11. **Identification:** The minimal witness proves erroneous complexity selection but does not independently identify a psychological probability-distortion mechanism.

8.3 Changes from the independent audit (29 October 2026)

1. Theorem 4.1 no longer assumes nesting, which Corollary 8.1 requires; nesting is used only for the likelihood-ratio statement.
2. Theorem 6.2 is proved from Theorem 7.2, which supplies the missing localization of the global projections; remainders are $o(\varepsilon^4)$, and $O(\varepsilon^5)$ for C^3 charts.
3. Appendix A gives complete proofs of small-cell exclusion, the evidence limit, uniform deletion concentration and the trial-LOO limit.
4. The proof of Theorem 7.2 applies the implicit-function theorem in the law variable and states the uniqueness argument.
5. Section 1 defines BIC, the evidence, the Bayes factor, ELPD/LOOIC and "selects".
6. $\theta_S^\star$ is corrected in the ninth digit to 0.4429045203.
7. Proposition 5.3 is stated in the δ coordinate and requires feasibility of the whole path segment.
8. C_M is defined for a chosen optimal law; Lemma 2.3's tautological sentence is replaced; π' and Ent are defined.
9. Remark 6.3 refers to positive laws; Proposition 7.5 states the reflection-invariance hypothesis its proof uses.
10. The witness uses the project's minimal-contract prior box; Proposition 7.7 uses the witness cell order (all values unchanged).
11. Remark 5.2 gives an explicit example with $\Lambda = 0$ and $\Delta_J > 0$; Proposition 7.4 also bounds Δ_C.
12. Change-log items on closure, drift, gap bound and higher moments now describe the original draft accurately.

8.4 Remaining research obligations

For a particular hierarchical G1 pair, the next rigorous task is a single *constructive* design/context law for which (i) $R_K > 0$; (ii) the joint advantage $\Delta_J > 0$; (iii) the separately computed conditional advantage $\Delta_C > 0$; and (iv) unique KL-optimal laws and prior support hold. An analytic or certified-numerical witness in an interior positive-variance regime would avoid the zero-variance geometry obstruction. A boundary-regime proof, if scientifically essential, should use tangent cones or directly controlled one-sided expansions. Continuous designs, $T \to \infty$, and local alternatives $\varepsilon_n \to 0$ require distinct asymptotic theorems and are not claimed here.

Reproducibility. The accompanying `verify_revised_theorem.py` independently recomputes the exact witness and the joint-versus-conditional sign-reversal diagnostics. The script `audit_revised_central_theorem.py`, in the project's `scripts` directory, recomputes every number displayed in this version, including the witness on the contract prior box, Proposition 7.7 in the witness cell order, the example of Remark 5.2 and the bound of Proposition 5.3, and tests Theorem 7.2 at an interior hierarchical baseline. Numerical checks support arithmetic, not the general proofs.

A Proof details for Theorem 4.1

Throughout, $j = (w, y)$ ranges over the $N = |\mathcal{W}|2^T$ cells, $r_j = H(w)p(y \mid w)$ and $r_{\min} = \min_j r_j > 0$. A candidate law is identified with its vector $q = (q_j)$, $q_j = q(y \mid w)$, in the finite product $\mathcal{S}$ of conditional simplices. "Positive prior mass on a law-neighborhood U" means $\Pi_M\{\phi : q_M(\cdot;\phi) \in U\} > 0$. For $u > 0$ let

$E_n(u) = \{\max_j |\hat{r}_j - r_j| \le u\}$; Hoeffding's inequality and a union bound give $\Pr\{E_n(u)^c\} \le 2Ne^{-2nu^2}$. For $a \in (0,1)$ let $B_a = \{q \in \mathcal{S} : \min_j q_j \ge a\}$. On $E_n(u)$, for every $q \in B_a$,

$$|\hat{\ell}_n(q) - \ell(q)| \le \log(1/a) \sum_j |\hat{r}_j - r_j| \le Nu \log(1/a). \tag{19}$$

Lemma A.1 (Small-cell exclusion and the likelihood limit)**.** *Fix M, an optimal law $q^\star_M \in \overline{\mathcal{Q}_M}$, and $a \in (0, \frac{1}{2}\min_j q^\star_{M,j})$ with $\frac{1}{2} r_{\min} \log a < \ell^\star_M - 2$. Let $u_0 = \min\{r_{\min}/2,\, 1/(N\log(1/a))\}$. On $E_n(u)$ with $u \le u_0$: (i) every $q \in \mathcal{Q}_M \setminus B_a$ has $\hat{\ell}_n(q) \le \ell^\star_M - 2$; (ii) $|\sup_{\mathcal{Q}_M} \hat{\ell}_n - \ell^\star_M| \le Nu\log(1/a)$. Consequently $\sup_{\mathcal{Q}_M} \hat{\ell}_n \to_p \ell^\star_M$ and $(\mathrm{BIC}_K - \mathrm{BIC}_S)/n \to_p -2\Delta_J$.*

Proof. On $E_n(u)$ every $\hat{r}_j \ge r_{\min}/2$. If $\min_j q_j < a$, every term of $\hat{\ell}_n(q)$ is nonpositive and one is at most $\frac{1}{2} r_{\min} \log a$, proving (i). Laws in $\mathcal{Q}_M$ near $q^\star_M$ have all cells at least a, and $\ell \le \ell^\star_M$ on $\mathcal{Q}_M$, so $\sup_{\mathcal{Q}_M \cap B_a} \ell = \ell^\star_M$ and (19) gives $|\sup_{\mathcal{Q}_M \cap B_a} \hat{\ell}_n - \ell^\star_M| \le Nu\log(1/a) \le 1$. By (i), laws outside B_a do not affect the supremum, proving (ii). The limit follows because $\Pr\{E_n(u)\} \to 1$ for each fixed u. Finally $(\mathrm{BIC}_K - \mathrm{BIC}_S)/n = -2(\sup_{\mathcal{Q}_K} \hat{\ell}_n - \sup_{\mathcal{Q}_S} \hat{\ell}_n) + (k_K - k_S) n^{-1} \log n$. □

Lemma A.2 (Evidence)**.** *Under the prior-support condition, $n^{-1}\log m_M(D) \to_p \ell^\star_M$; hence $n^{-1}\log \mathrm{BF}_{KS} \to_p \Delta_J$.*

Proof. Since Π_M is a probability measure, $m_M(D) \le \exp\{n \sup_{\mathcal{Q}_M} \hat{\ell}_n\}$. For $\delta \in (0,1)$, the set $U_\delta = \{q \in \mathcal{S} : \ell(q) > \ell^\star_M - \delta,\ \min_j q_j > a\}$ is open, because ℓ is continuous on positive laws, and contains $q^\star_M$; hence $\pi_\delta = \Pi_M\{\phi : q_M(\cdot;\phi) \in U_\delta\} > 0$. On $E_n(u)$, (19) gives $m_M(D) \ge \pi_\delta \exp\{n(\ell^\star_M - \delta - Nu\log(1/a))\}$. Given $\eta > 0$, take $\delta = \eta/3$ and u with $Nu\log(1/a) \le \eta/3$; for n so large that $n^{-1}\log \pi_\delta \ge -\eta/3$, $n^{-1}\log m_M(D) \ge \ell^\star_M - \eta$ on $E_n(u)$. Combine with the upper bound and Lemma A.1. □

Lemma A.3 (Posterior concentration, uniformly over deletions)**.** *Suppose $\overline{\mathcal{Q}_M}$ has a unique optimal law $q^\star_M$, and let U be an open neighborhood of it in $\mathcal{S}$. Then*

$$\max_{1 \le i \le n} \Pi_M(q \notin U \mid D_{-i}) \xrightarrow{p} 0, \qquad \max_{i,t} \Pi_M(q \notin U \mid D_{-it}) \xrightarrow{p} 0,$$

where D_{-i} deletes participant i and D_{-it} deletes only y_{it}.

Proof. Shrinking U only strengthens the conclusion, so assume $U \subseteq B_{2a}$ with a as in Lemma A.1. The set $(\overline{\mathcal{Q}_M} \cap B_a) \setminus U$ is compact; ℓ is continuous on it and, by uniqueness, strictly below $\ell^\star_M$ there. Choose $g \in (0,1]$ with $\ell \le \ell^\star_M - g$ on that set, and let $V = U \cap \{q : \ell(q) > \ell^\star_M - g/4\}$, an open neighborhood of $q^\star_M$ with $\pi_V = \Pi_M\{\phi : q_M(\cdot;\phi) \in V\} > 0$.

Participant deletion. The frequencies without participant i are $\hat{r}^{(-i)} = (n\hat{r} - \mathbf{1}_{j_i})/(n-1)$, so $\max_{i,j} |\hat{r}^{(-i)}_j - \hat{r}_j| \le 1/(n-1)$. On $E_n(u)$, every $\hat{r}^{(-i)}$ is therefore within $u' = u + 1/(n-1)$ of r, simultaneously for all i. Take u small and n large so that $u' \le u_0$ and $Nu'\log(1/a) \le g/8$. Applying Lemma A.1(i) and (19) to the deleted frequencies, the deleted log likelihood $(n-1)\hat{\ell}^{(-i)}_n(q)$ is at most $(n-1)(\ell^\star_M - 7g/8)$ for $q \notin U$ and at least $(n-1)(\ell^\star_M - 3g/8)$ on V. Hence, on the single event $E_n(u)$,

$$\max_i \Pi_M(q \notin U \mid D_{-i}) \le \pi_V^{-1} e^{-(n-1)g/2}.$$

Response deletion. $\Pi_M(d\phi \mid D_{-it}) \propto q_{M,-t}(y_{i,-t} \mid W_i; \phi)\, \Pi_M(d\phi \mid D_{-i})$, where $q_{M,-t}$ is the marginal law of the retained responses. The tilt is at most one, and on $U \subseteq B_{2a}$ it is at least $2a$, since every marginal cell of a law in B_{2a} is at least $2a$. Therefore

$$\Pi_M(q \notin U \mid D_{-it}) \le \frac{\Pi_M(q \notin U \mid D_{-i})}{2a\, \Pi_M(q \in U \mid D_{-i})},$$

and the right side tends to zero uniformly in (i, t) by the first part. □

Lemma A.4 (Trial-LOO limit)**.** *If both models satisfy the hypotheses of Lemma A.3, then* $(\mathrm{ELPD}_K - \mathrm{ELPD}_S)/(nT) \to_p \Delta_C$.

Proof. For a positive law q and a cell (w, y) let $F_t(q; w, y) = q(y \mid w)/q_{-t}(y_{-t} \mid w) = q(y_t \mid y_{-t}, w) \in (0, 1]$. By (13), $\widehat{p}_M(y_{it} \mid D_{-it}) = \int F_t(q_M(\cdot; \phi); W_i, Y_i)\, \Pi_M(d\phi \mid D_{-it})$. The finitely many maps $F_t(\cdot; w, y)$ are uniformly continuous on B_a. Let $q^\star_{\min} = \min_j q^\star_{M,j}$ and $\eta \in (0, q^\star_{\min}/4]$, and choose a neighborhood $U \subseteq B_{2a}$ of $q^\star_M$ on which $|F_t(q; w, y) - F_t(q^\star_M; w, y)| \le \eta$ for all (t, w, y). Since $0 \le F_t \le 1$,

$$\max_{i,t} \left|\widehat{p}_M(y_{it} \mid D_{-it}) - F_t(q^\star_M; W_i, Y_i)\right| \le \eta + \max_{i,t} \Pi_M(q \notin U \mid D_{-it}),$$

which is at most 2η with probability tending to one by Lemma A.3. Because $F_t(q^\star_M; \cdot) \ge q^\star_{\min}$, all predictions then exceed $q^\star_{\min}/2$, where $\log$ is Lipschitz with constant $2/q^\star_{\min}$. Hence

$$\frac{\mathrm{ELPD}_M}{nT} = \sum_j \hat{r}_j\, \mathcal{H}_M(j) + o_p(1), \qquad \mathcal{H}_M(w, y) = \frac{1}{T} \sum_{t=1}^{T} \log q^\star_M(y_t \mid y_{-t}, w),$$

and $\sum_j \hat{r}_j \mathcal{H}_M(j) \to_p \sum_j r_j \mathcal{H}_M(j)$ because $\mathcal{H}_M$ is bounded. Finally $\sum_j r_j\{\mathcal{H}_K(j) - \mathcal{H}_S(j)\} = C_S - C_K = \Delta_C$, since the true conditional log-probabilities cancel. The law of large numbers is applied to participant cells; no independence between a participant's trials is used. □